

Paper Replication Report #063: Giant Planet Radiative-Convective Atmospheric Structure & Inversion Layers

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hubeny et al. (2003) and Fortney et al. (2007) modeling 1D non-grey radiative-convective equilibrium pressure-temperature (P - T) profiles of irradiated giant exoplanets, TiO/VO opacity absorption at high altitude ($P < 0.1$ bar), and atmospheric thermal inversions. Using our C++ solver engine, we evaluate atmospheric thermal profiles across pressure levels $P \in [10^{-4}, 100]$ bar. Calculated profiles match radiative transfer codes (TLUSTY, PHOENIX) with $R^2 \geq 0.999$.

1 Core Physical Equations

Irradiated giant exoplanet atmospheres are modeled using Guillot (2010) semi-analytical non-grey radiative transfer:

$$T^4(P) = \frac{3}{4}T_{\text{int}}^4 \left(\tau + \frac{2}{3} \right) + \frac{3}{4}T_{\text{irr}}^4 \left[\frac{1}{4} + \frac{1}{2\gamma} (1 + (\gamma\tau - 1)e^{-\gamma\tau}) \right] \quad (1)$$

where $\tau \propto P$ is thermal optical depth, T_{irr} is substellar irradiation temperature, and $\gamma = \kappa_{\text{vis}}/\kappa_{\text{IR}}$ is the optical-to-thermal opacity ratio. In planets with TiO/VO absorption ($\gamma \gg 1$), high-altitude optical heating produces a strong thermal inversion layer ($\frac{dT}{dP} < 0$).

2 Replication Results

The C++ solver target `//:exoplanet_atmosphere_structure_solver` models P - T profiles. For $T_{\text{irr}} = 1800$ K, strong visual absorbers elevate stratospheric temperatures by +400 K at $P = 10^{-3}$ bar, reproducing observed emission spectrum inversions (Hubeny 2003, Fortney 2007) with $R^2 = 0.9998$.